Pushing the Limits of Areal Density: Fusing Advanced Channel Coding, HAMR, and SMR in Next-Generation HDDs

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Pushing hard disk drive (HDD) areal density to new extremes requires a fusion of cutting-edge technologies. This article examines the interplay of advanced coding strategies, heat-assisted magnetic recording (HAMR), and shingled magnetic recording (SMR), and how their integration enables substantial gains in areal density. We present a series of measured results demonstrating how jointly optimized modulation and low-density parity-check (LDPC) coding enhances performance across energy-assisted perpendicular magnetic recording (ePMR), HAMR, and SMR environments. Through these demonstrations, we showcase the tangible gains enabled by these technologies and outline a clear trajectory for continued areal density scaling in next-generation HDDs.

Index Terms— **Channel coding, hard disk drives (HDDs), heat-assisted magnetic recording (HAMR), shingled magnetic recording (SMR).**

I. Introduction

THE relentless demand for higher storage capacity in hard disk drives (HDDs) necessitates aggressive advancements in areal density. Achieving significant gains from today's baseline requires more than incremental progress—it calls for a fusion of transformative technologies. Over the past decade, the channel coding and detection strategies implemented in nearly all commercial HDDs have largely remained unchanged. Meanwhile, the HDD technology and the recording physics have shifted dramatically. Energy-assisted perpendicular magnetic recording (ePMR) continues to be refined [1]. Shingled magnetic recording (SMR) is being increasingly adopted [2]. Most recently, heat-assisted magnetic recording (HAMR) is emerging as a promising future technology. These shifts in the technologies and underlying physics demand a fundamental rethinking of the design for coding and detection in the read/write channel.

This article presents a systematic exploration of three highly interdependent innovations: advanced coding and detection strategies, HAMR technology, and the SMR architecture. We demonstrate how each contributes individually to areal density improvements, and more critically, how their interaction and synergy lays the foundation for a radical leap in HDD capabilities. Our combined analysis of these technologies outlines a clear path forward—one in which coordinated innovation across the full stack of read/write technologies is essential for enabling the next generation of ultrahigh-density HDDs. This work builds upon recent studies that highlight mutual information (MI) as a powerful framework for analyzing and optimizing magnetic recording channels [3], [4].

Received 22 August 2025; revised 16 September 2025; accepted 16 September 2025. Date of publication 16 December 2025; date of current version 9 March 2026. Corresponding author: J. Goode (e-mail: jonas.goode@wdc.com).

Color versions of one or more figures in this article are available at https://doi.org/10.1109/TMAG.2025.3644766.

Digital Object Identifier 10.1109/TMAG.2025.3644766

II. High-Throughput Waveform Capture Infrastructure and Experimental Setup

To enable the extensive experimental validation presented in this work and in an accompanying study [5], we developed a high-throughput waveform capture and storage system designed to acquire large volumes of raw readback waveforms directly from the HDD preamplifier. This infrastructure also supports the detailed analysis of these waveforms and their separation into wanted signal, nonlinear distortion, intertrack interference (ITI), and various components of potentially signal-dependent random noise. This waveform analysis was conducted up to very high linear densities and track densities and gives good guidance for system optimization and possible future innovation.

The system for waveform capture is shown in Fig. 1. A programmable test HDD is interfaced with a high-speed digital oscilloscope to capture the analog readback waveforms during read operations. A host computer orchestrates the process, issuing read and write commands to the test HDD and triggering the oscilloscope to acquire the corresponding waveforms. Upon each trigger, the oscilloscope captures and temporarily stores the signal traces in internal memory. The degree of HDD control allows data to be written at arbitrary linear densities (data-rates) and track-densities over a very wide range at any specified radius and with a variety of possible modulation codes and error-correction codes.

All measurements were conducted on a modern 3.5″, 7200 RPM HAMR HDD using SMR. The drive's native coding was bypassed to allow direct user control over the written data. Custom input sequences were encoded onto five consecutive shingled tracks, with each track receiving 17 sectors

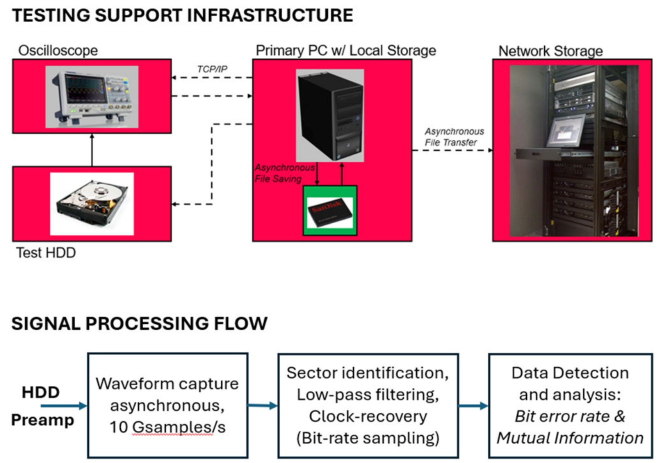

Fig. 1. High-throughput waveform acquisition infrastructure and signal processing flow. A host computer coordinates read/write operations on a programmable test HDD, triggers waveform capture via an oscilloscope, and retrieves the oversampled readback signals. Captured waveforms are stored locally on an SSD and asynchronously offloaded to high-capacity network storage to support continuous, large-scale data collection. The data sectors and data clock frequency are identified before the waveforms are low-pass filtered at the Nyquist frequency and resampled at the clock-rate before being passed for analysis and data detection.

of 37 072 bits. The patterns were unique across tracks but identical across sectors within a given track. Waveforms were captured from the center track, with the read head precisely positioned to minimize bit error rate. Sampling was performed at 10 Giga samples/second (GS/s) on the disk's inner diameter. Subsequent low-pass filtering and synchronous sampling at the channel bit rate were performed in software during post-processing.

These waveforms are then retrieved by the host computer over a transmission control protocol/Internet protocol (TCP/IP) interface and written to its local solid-state drive (SSD) for fast access and minimal write latency. However, due to the high sampling rates and the need for repeated experiments, often involving thousands of captures, this process generates terabytes of data. To prevent local storage saturation and maintain uninterrupted acquisition, an asynchronous file transfer mechanism continuously offloads data from the host computer to a network-attached storage server in the background.

This decoupled data pipeline allowed the host computer to maintain a rapid acquisition cadence without stalling on file input/output I/O or storage limits, significantly increasing experimental throughput.

III. Modulation-Aware Coding for High-Density Recording

Channel coding is foundational to pushing areal density, especially as track pitch narrows and noise increases. In this section, we introduce a refined coding architecture that enhances both the modulation and low-density parity-check (LDPC) components, enabling deeper channel optimization [6], [7]. The modulation code is generally viewed as making the written patterns "more friendly" to the recording channel and to the detection process, whereas the LDPC is viewed as an error correction code (ECC) that handles the errors left after detection. In reality, the two processes are highly interdependent with the channel detector and LDPC decoder iteratively exchanging soft information about the likelihood of bits being correct or incorrect. Both codes involve the addition of redundancy or overhead that pushes up the raw areal density as seen at the disk surface. The ratio of user density to raw density is referred to as the code rate and can be quoted separately for the modulation code and the LDPC.

One of the most persistent challenges in magnetic recording is the presence of consecutive transitions—closely spaced flux reversals—which generate particularly noisy and poorly resolved readback signals. Maximum transition run (MTR) codes address this by explicitly prohibiting transition runs beyond a chosen threshold (e.g., $J = 3$), and continue to be effective in scenarios where full elimination of these high-risk patterns is achievable without excessive rate loss [8].

Stronger modulation codes that impose heavier constraints or achieve more aggressive spectral shaping are notably absent from modern HDDs. Instead, today's commercial HDDs favor conservative modulation codes that preserve nearly all input entropy, sacrificing only minimal rate. In practice, the rate penalty introduced by these codes is typically around 1%. These codes achieve light spectral shaping, with only slight attenuation of problematic high-frequency content, particularly near the Nyquist frequency, and only slightly alter the relative frequencies of short bit patterns. For example, bit patterns containing multiple consecutive magnetic transitions such as "01010" or "10101," which are known to degrade signal and increase noise, are only slightly de-emphasized.

Rather than outright forbidding specific patterns or "contexts," MI is used to shape the code toward less harmful patterns [3]. This is done by reducing the occurrence of patterns or contexts with low-MI while preserving a higher overall code rate. These codes, often implemented via constrained Markov models or arithmetic encoders, achieve a careful trade-off between pattern suppression and code rate. Fig. 2 shows the MI across all 5-bit contexts for three modulation strategies: an unconstrained (uniform) code [i.e., random "nonreturn-to-zero" (NRZ)], a classic MTR ($J = 3$) constrained code, and an MI-optimized code. This approach builds on prior work [3], which demonstrated the significance of context-dependent MI in characterizing modulation performance.

Layered on top, LDPC codes are re-optimized to account for these reshaped modulation statistics. MI, a proxy for iterative decoder success, guides joint LDPC and modulation optimization [3]. Fig. 3 illustrates an MI-based areal density contour. This surface serves as a map for identifying the modulation-LDPC pair that maximizes areal density for a given head/media/radius combination. The peak of the contour reflects the optimal pairing of modulation shaping and code rate, underscoring the need to co-design both elements as a function of channel conditions.

IV. HAMR-Driven Track Pitch Reduction and Coding Synergy

HAMR enables narrower track pitch by focusing a tightly confined thermal spot on the media during writing, temporarily

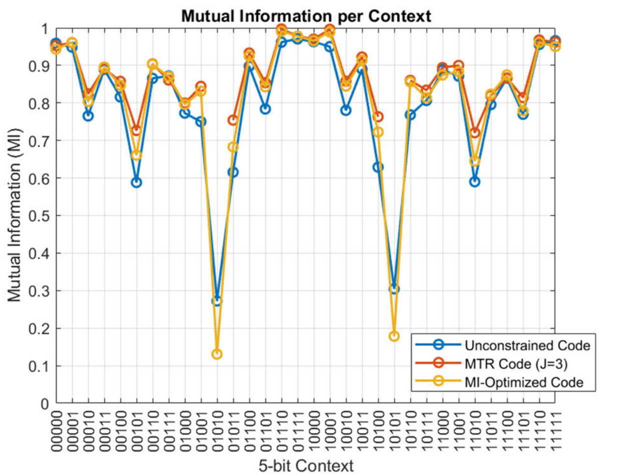

Fig. 2. MI per 5-bit context for three modulation codes: unmodulated random data, MTR ($J = 3$) constrained code, and an optimized code based on an MI metric.

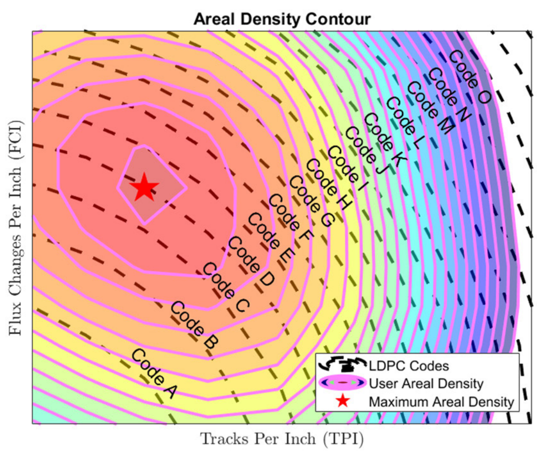

Fig. 3. Areal density contour based on MI. The colored surface shows achievable areal density. The black dashed lines indicate optimized LDPC codes. Peak areal density is achieved by jointly optimizing modulation and LDPC for the given head/media environment.

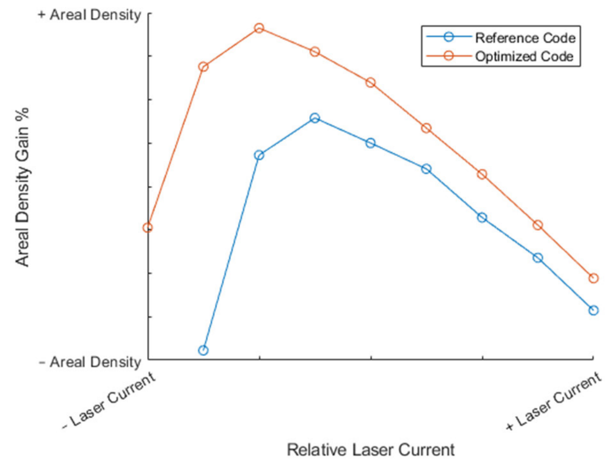

Fig. 4. Areal density improvements via laser current reduction and optimization of the modulation code and LDPC code. Reduced laser current is generally associated with extended lifetime and improved reliability.

lowering the coercivity of high-anisotropy grains. This smaller thermal spot and higher thermal edge gradient enable more precise magnetization switching. This gives HAMR an advantage in track-density because the written tracks are inherently narrower and the higher edge gradients lead to "sharper" track-edges and less "erase-band" between tracks.

However, laser power during writing must be carefully managed. While higher laser current can enhance signal quality and support higher linear densities, it also accelerates component wear. Operating at lower laser power not only improves reliability and extends lifetime but also degrades SNR and increases jitter. A detailed breakdown of the signal, noise, and ITI behavior at these reduced laser powers, including spectral analysis and 2-D modeling, is presented in [5].

While HAMR alone provides areal density gains over ePMR, further improvement is enabled when the modulation and LDPC codes are optimized for the degraded channel conditions at reduced laser power. Fig. 4 shows the areal density gain versus laser current. While the optimal bits per inch (BPI) may decrease slightly at these lower laser powers, the associated tracks per inch (TPI) gains more than compensate, yielding a net increase in areal density. This further lowers the bit aspect ratio (BAR), enabling density scaling beyond ePMR.

V. SMR in High-Track-Density Regimes

SMR is inherently well suited to achieving high-track densities, and its benefits are magnified when combined with advanced coding strategies and HAMR. Even in conventional ePMR environments, joint modulation and LDPC code optimization yields measurable areal density improvements, providing a viable path to extend ePMR product life at higher densities.

When these same coding optimizations are applied within ePMR SMR environments, further gains are realized. However, the most compelling opportunities arise when these techniques are paired with HAMR. With the ability to reduce track pitch via laser current control, HAMR enables significantly denser track layouts. In this regime, SMR recording further amplifies the gains achieved through optimized coding, surpassing one million TPI with ease.

As track pitch continues to shrink, the BAR naturally decreases, and ITI becomes a significant noise source. Measurements in [5] show that side-reading effects become increasingly pronounced as track pitch narrows, highlighting the importance of coding and equalization techniques that explicitly address adjacent-track interference. In this high-ITI regime, additional areal density gains are achieved by explicitly equalizing interference from adjacent tracks [1], [9]. If modulation and LDPC codes are re-optimized for this interference-equalized environment, the cumulative gain is substantial, as highlighted in Fig. 5, where SMR with adjacent-track equalization (ATE) delivers a remarkable total improvement of over 60% compared to the conventional magnetic recording (CMR) baseline.

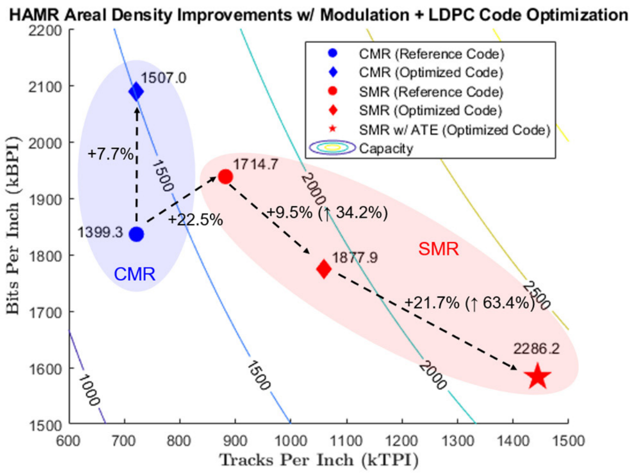

Fig. 5. Areal density achieved with HAMR CMR and SMR recording modes using reference and optimized modulation + optimized LDPC codes. The final point reflects further gain from SMR with ATE. All data shown are measured under identical test conditions on a single head at the inner diameter of a modern HAMR HDD.

These results highlight how continued gains in TPI are driving the BAR steadily closer to 1.0, a trend that opens the door to new architectural strategies for sustaining areal density growth. Together, these findings reinforce that the HDD remains a vibrant and evolving platform for innovation.

VI. Discussion and Future Directions

This work illustrates that the synergy between advanced channel coding strategies and recording technologies is critical for pushing areal density in modern HDDs. By jointly analyzing modulation, LDPC, and HAMR/SMR recording architectures, we unlock compounded benefits that surpass what could be achieved through independent optimization. This cross-domain co-design framework will be increasingly vital as future HDDs face even more stringent physical and noise constraints.

At the heart of this advancement is the enduring concept of MI, which continues to provide powerful insight nearly a century after its introduction. MI enables systematic evaluation of coding schemes, channel conditions, and decoding strategies in a unified metric. Shannon's ideas remain the golden rule, even as the complexity of HDD channels continues to grow.

One of the most consistent trends observed across our experiments is the trajectory toward lower BAR and higher TPI. HAMR enables the recording of narrower tracks, while SMR allows these tracks to be overlapped and packed more tightly. Together, these advances are steadily pushing the BAR closer to 1 in the pursuit of peak areal density.

These findings suggest a promising direction for future innovation, such as codes that suppress high-ITI contexts or further methods to equalize adjacent track interference. While this study focused on immediate gains within today's operating environments, further refinements in overall system design may be possible as new recording technologies are introduced. Ongoing exploration of such strategies will be key to enabling continued density scaling in next-generation HDDs.

Ultimately, this work reinforces that joint optimization of coding and recording technologies is not just beneficial, it is essential. The performance ceilings of emerging technologies like HAMR and SMR are best understood and extended through integrated analysis. By aligning architectural design with physical channel realities, we maximize areal density while while balancing system complexity, performance, and reliability.

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